

## The matrix of change of basis

$T_{BB'}$  → on columns we write the vectors in  $B$  as lin. comb. of vectors in  $B'$ .

$$T_{BB'} = T_{B'B}^{-1}$$

$$[u]_B = T_{BB'} \cdot [u]_{B'}$$

$$\begin{aligned} f \text{ endomorphism} \Rightarrow [f]_B &= T_{BB'} \cdot [f]_{B'} \cdot T_{B'B} \\ &= T_{B'B}^{-1} \cdot [f]_{B'} \cdot T_{B'B} \end{aligned}$$

## The characteristic polynomial:

$$p(\lambda) = \det(\underbrace{[f]_E - \lambda I_n}_A)$$

Eigenvalues:  $p(\lambda) = 0 \Rightarrow$  solutions...

Eigenvectors:

$$A \cdot \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = 0_n.$$

①  $\mathbb{R}^3$ .

$B = (v_1, v_2, v_3) = ((1, 0, 1), (0, 1, 1), (1, 1, 1))$  basis

$B' = (v'_1, v'_2, v'_3) = ((1, 1, 0), (-1, 0, 0), (0, 0, 1))$  basis.

$$T_{BB'} = ?$$

$$T_{B'B} = ?$$

Compute coord. of  $u = (2, 0, -1)$  in  $B$  &  $B'$ .

$$v_1 = \alpha v'_1 + \beta v'_2 + \gamma v'_3$$

$$(1, 0, 1) = \alpha(1, 1, 0) + \beta(-1, 0, 0) + \gamma(0, 0, 1)$$

$$(1, 0, 1) = (\alpha - \beta, \alpha, \gamma)$$

$$\Rightarrow \begin{cases} \alpha - \beta = 1 \\ \alpha = 0 \\ \gamma = 1 \end{cases} \Rightarrow \begin{cases} \beta = -1 \\ \alpha = 0 \\ \gamma = 1 \end{cases}$$

$$v_2 = (\alpha - \beta, \alpha, \gamma)$$

$$(0, 1, 1) = (\alpha - \beta, \alpha, \gamma)$$

$$\Rightarrow \begin{cases} \alpha - \beta = 0 \\ \alpha = 1 \\ \gamma = 1 \end{cases} \Rightarrow \begin{cases} \beta = 1 \\ \alpha = 1 \\ \gamma = 1 \end{cases}$$

$$\Rightarrow T_{B'B} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{cases} \alpha = 1 \\ \gamma = 1 \end{cases} \quad \begin{cases} \alpha = 1 \\ \gamma = 1 \end{cases}$$

$$T_{BB}^{-1} = \begin{bmatrix} -1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$N_3 = (\alpha - \beta, \alpha, \gamma)$$

$$\Rightarrow (1, 1, 1) = (\alpha - \beta, \alpha, \gamma)$$

$$\Rightarrow \begin{cases} \alpha - \beta = 1 \\ \alpha = 1 \\ \gamma = 1 \end{cases} \Rightarrow \begin{cases} \beta = 0 \\ \alpha = 1 \\ \gamma = 1 \end{cases}$$

$$T_{B'B} \rightarrow T_{BB'}^{-1} = \frac{1}{\det T_{B'B}} \cdot T_{B'B}^* \rightarrow \text{the inverse could also be computed in another way.}$$

or

$$N_1' = \alpha N_1 + \beta N_2 + \gamma N_3, \dots$$

$$\Rightarrow T_{BB'}^{-1} = -1 \cdot \begin{bmatrix} 1 & 0 & -1 \\ 1 & -1 & -1 \\ -2 & 1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 1 & 1 \\ 2 & -1 & -1 \end{bmatrix}$$

$$u = (2, 0, -1) = \alpha_1 N_1 + \alpha_2 N_2 + \alpha_3 N_3$$

$$= \alpha_1 (1, 0, 1) + \alpha_2 (0, 1, 1) + \alpha_3 (1, 1, 1)$$

$$= (\alpha_1 + \alpha_3, \alpha_2 + \alpha_3, \alpha_1 + \alpha_2 + \alpha_3)$$

$$\Rightarrow \begin{cases} \alpha_1 + \alpha_3 = 2 \\ \alpha_2 + \alpha_3 = 0 \\ \alpha_1 + \alpha_2 + \alpha_3 = -1 \end{cases} \Rightarrow \begin{cases} \alpha_1 - \alpha_2 = 2 \\ \alpha_3 = -\alpha_2 \end{cases} \Rightarrow \alpha_1 = 2 + \alpha_2$$

$$2 + \alpha_2 + \alpha_2 - \alpha_2 = -1 \Rightarrow \alpha_2 = -3$$

$$\alpha_3 = 3$$

$$\alpha_1 = -1$$

$$\Rightarrow [u]_B = \begin{bmatrix} -1 \\ -3 \\ 3 \end{bmatrix}$$

$$u = (2, 0, -1) = \alpha_1 N_1' + \alpha_2 N_2' + \alpha_3 N_3'$$

$$= \alpha_1 (1, 1, 0) + \alpha_2 (-1, 0, 0) + \alpha_3 (0, 0, 1)$$

$$\Rightarrow \begin{cases} \alpha_1 - \alpha_2 = 2 \\ \alpha_1 = 0 \\ \alpha_3 = -1 \end{cases} \Rightarrow \begin{cases} \alpha_2 = -2 \\ \alpha_1 = 0 \\ \alpha_3 = -1 \end{cases} \Rightarrow [u]_{B'} = \begin{bmatrix} 0 \\ -2 \\ -1 \end{bmatrix}$$

or by using  $[u]_B = T_{BB'} \cdot [u]_{B'}$

Check  $T_{BB'} \cdot T_{B'B} = I_n!$

②  $\mathbb{R}^2$

$$B = (v_1, v_2) = ((1, 2), (1, 3))$$

$$B' = (v_1', v_2') = ((1,0), (2,1))$$

$$f, g \in \text{End}_{\mathbb{R}}(\mathbb{R}^2)$$

$$[f]_B = \begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix}$$

$$[g]_{B'} = \begin{bmatrix} -7 & -13 \\ 5 & 7 \end{bmatrix}$$

|                                                |
|------------------------------------------------|
| $[g]_B = T_{BB'} \cdot [g]_{B'} \cdot T_{B'B}$ |
| $[f]_{B'} = T_{BB'} \cdot [f]_B \cdot T_{B'B}$ |

$$[f+g]_B = [f]_B + [g]_B \rightarrow \text{I need } [g]_B$$

$$[f \circ g]_{B'} = [f]_B \cdot [g]_{B'} \rightarrow \text{I need } [f]_{B'}$$

$$v_1 = \alpha v_1' + \beta v_2'$$

$$(1,2) = \alpha(1,0) + \beta(2,1)$$

$$(1,2) = (\alpha + 2\beta, \beta)$$

$$\Rightarrow \begin{cases} \alpha + 2\beta = 1 \\ \beta = 2 \end{cases} \Rightarrow \begin{aligned} \alpha + 4 &= 1 \\ \alpha &= -3 \\ \beta &= 2 \end{aligned}$$

$$\Rightarrow T_{BB'} = \begin{bmatrix} -3 & -5 \\ 2 & 3 \end{bmatrix}$$

$$v_2 = \alpha v_1' + \beta v_2'$$

$$\Rightarrow \begin{cases} \alpha + 2\beta = 1 \\ \beta = 3 \end{cases} \Rightarrow \begin{aligned} \alpha + 6 &= 1 \\ \alpha &= -5 \\ \beta &= 3 \end{aligned}$$

$$v_2' = \alpha v_1 + \beta v_2$$

$$(2,1) = (\alpha + \beta, 2\alpha + 3\beta)$$

$$\Rightarrow \begin{cases} \alpha + \beta = 2 \\ 2\alpha + 3\beta = 1 \end{cases} \Rightarrow \begin{aligned} \alpha &= 2 - \beta \\ 2(2 - \beta) + 3\beta &= 1 \\ 4 - 2\beta + 3\beta &= 1 \\ 4 + \beta &= 1 \end{aligned}$$

$$\begin{aligned} \beta &= -3 \\ \alpha &= 5 \end{aligned}$$

$$\Rightarrow T_{B'B} = \begin{bmatrix} 3 & 5 \\ -2 & -3 \end{bmatrix}$$

$$v_1' = \alpha v_1 + \beta v_2$$

$$\Rightarrow \begin{cases} \alpha + \beta = 1 \\ 2\alpha + 3\beta = 0 \end{cases} \Rightarrow \begin{aligned} \alpha &= 1 - \beta \\ 2(1 - \beta) + 3\beta &= 0 \\ 2 - 2\beta + 3\beta &= 0 \\ \beta + 2 &= 0 \\ \beta &= -2 \end{aligned}$$

$$\lambda = 3$$

$$[g]_B = T_{BB'} \cdot [g]_{B'} \cdot T_{B'B}$$

$$[g]_B = \begin{bmatrix} -3 & -5 \\ 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} -7 & -13 \\ 5 & 7 \end{bmatrix} \cdot \begin{bmatrix} 3 & 5 \\ -2 & -3 \end{bmatrix} \dots$$

$$= \begin{bmatrix} -20 & -32 \\ 13 & 20 \end{bmatrix}$$

$$[f]_{B'} = T_{BB'} \cdot [f]_B \cdot T_{B'B}$$

$$[f]_{B'} = \begin{bmatrix} 3 & -5 \\ 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix} \cdot \begin{bmatrix} 3 & 5 \\ -2 & -3 \end{bmatrix} = \dots$$

$$= \begin{bmatrix} 8 & 13 \\ -5 & -8 \end{bmatrix}$$

$$[f+g]_B = [f]_B + [g]_B = \begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix} + \begin{bmatrix} -20 & -32 \\ 13 & 20 \end{bmatrix} = \begin{bmatrix} -19 & -30 \\ 12 & 19 \end{bmatrix}$$

$$[f \circ g]_{B'} = [f]_{B'} \cdot [g]_{B'} = \begin{bmatrix} 2 & 7 \\ -5 & -8 \end{bmatrix} \cdot \begin{bmatrix} -7 & -13 \\ 5 & 7 \end{bmatrix} = \begin{bmatrix} 21 & 23 \\ -5 & 9 \end{bmatrix}$$

$$\textcircled{4} f \in \text{End}_{\mathbb{R}}(\mathbb{R}^2)$$

$$f(x, y) = (3x + 3y, 2x + 4y)$$

a) eigenvalues, eigenvectors of  $f$ .

$$[f]_E = \begin{bmatrix} 3 & 3 \\ 2 & 4 \end{bmatrix}$$

$$p(\lambda) = \det([f]_E - \lambda I_m) = 0$$

$$A \cdot \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix} = 0_m$$

$$0 = p(\lambda) = \begin{vmatrix} 3-\lambda & 3 \\ 2 & 4-\lambda \end{vmatrix} = (3-\lambda)(4-\lambda) - 6 = 12 - 7\lambda + \lambda^2 - 6$$

$$= \lambda^2 - 7\lambda + 6 = \lambda^2 - \lambda - 6\lambda + 6$$

$$= \lambda(\lambda-1) - 6(\lambda-1)$$

$$0 = (\lambda-1)(\lambda-6) \rightarrow \begin{matrix} \lambda_1 = 1 \\ \lambda_2 = 6 \end{matrix} \left. \vphantom{\begin{matrix} \lambda_1 = 1 \\ \lambda_2 = 6 \end{matrix}} \right\} \text{eigenvalues.}$$

$$\begin{bmatrix} 3-\lambda & 3 \\ 2 & 4-\lambda \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{cases} (3-\lambda)x_1 + 3x_2 = 0 \\ 2x_1 + (4-\lambda)x_2 = 0 \end{cases}$$

Case I:  $\lambda_1 = 1$ .

$$\Rightarrow \begin{cases} 2x_1 + 3x_2 = 0 \\ 2x_1 + 3x_2 = 0 \end{cases} \Rightarrow x_1 = -\frac{3}{2}x_2 = -\frac{3}{2}\alpha$$

$$\Rightarrow v_1 = \begin{bmatrix} -\frac{3}{2}\alpha \\ \alpha \end{bmatrix} \Rightarrow v_1(\lambda_1) = \langle \left(-\frac{3}{2}, 1\right) \rangle$$

Case II:  $\lambda_2 = 6$

$$\Rightarrow \begin{cases} -3x_1 + 3x_2 = 0 \\ 2x_1 - 2x_2 = 0 \end{cases} \Rightarrow \begin{cases} -x_1 + x_2 = 0 \\ -x_1 + x_2 = 0 \end{cases} \Rightarrow x_2 = x_1 = \alpha$$

$$\Rightarrow v_2 = \begin{bmatrix} \alpha \\ \alpha \end{bmatrix} \xrightarrow{\alpha=1} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \Rightarrow v_2(\lambda_2) = \langle (1, 1) \rangle$$

Lin. indep?  $\begin{vmatrix} 1 & 1 \\ -\frac{3}{2} & 1 \end{vmatrix} = 1 + \frac{3}{2} = \frac{5}{2} \neq 0$

$$\Rightarrow \nexists \alpha \text{ s.t. } v(\lambda_2) = \alpha \cdot v(\lambda_1)$$

$$\Rightarrow B = \left\langle (1, 1), \left(-\frac{3}{2}, 1\right) \right\rangle.$$

(9)  $A \in M_2(\mathbb{R})$

$\lambda_1, \lambda_2$  eigenvalues of  $A$  in  $\mathbb{C}$ .

a)  $\lambda_1 + \lambda_2 = \text{Tr}(A)$

b)  $p(\lambda) = \det(A - \lambda I_2) = \begin{vmatrix} a-\lambda & b \\ c & d-\lambda \end{vmatrix} = (a-\lambda)(d-\lambda) - bc$   
 $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$   
 $= \lambda^2 - \lambda(a+d) + \underline{ad - bc}$   
 $= \lambda^2 - \lambda \cdot \text{Tr } A + \underline{\det(A)} \quad (1)$

$$p(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2) = \lambda^2 - \lambda\lambda_2 - \lambda_1\lambda + \lambda_1\lambda_2 = \lambda^2 - \lambda(\lambda_1 + \lambda_2) + \lambda_1\lambda_2$$

$$\Rightarrow \begin{cases} \lambda_1 + \lambda_2 = \text{Tr}(A) = a + d \\ \lambda_1 \cdot \lambda_2 = \det(A) = ad - bc. \end{cases}$$

Exam! Prove ... using that  $\text{Tr}(A) = \lambda_1 + \lambda_2$  and  $\det(A) = ad - bc$ .

c) A has all eigenvalues in  $\mathbb{R} \Leftrightarrow (\text{Tr}(A))^2 - 4 \det(A) \geq 0$  ?

$$p(\lambda) \stackrel{(1)}{=} \lambda^2 - \lambda \cdot \text{Tr}(A) + \det(A)$$

$$p(\lambda) = 0 \Rightarrow \Delta = (-\text{Tr}(A))^2 - 4 \cdot \det(A)$$

If  $\Delta \geq 0 \Rightarrow \exists \lambda_1, \lambda_2 \in \mathbb{R}$ , not necessarily equal (eigenvalues)

(10.) Hint :  $\det(A + iI_2) = 0 = p(\lambda)$

$\det(A + 2iI_2) = ? \rightarrow \lambda = -i$  (eigenvalue)

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$p(\lambda) = \det([f]_E - \lambda \cdot I_m)$$

$$p(i) = \det(A - (-i) \cdot I_2)$$

$$\begin{cases} p(i) = (i - \lambda_1)(i - \lambda_2) & (1) \end{cases}$$

$$\begin{cases} p(i) = i^2 - i \cdot (a+d) + \underline{ad - bc} = -1 - i \cdot \text{Tr}(A) + \det(A) & (2) \end{cases}$$

$$(1) = (2) = 0 \Rightarrow (i - \lambda_1)(i - \lambda_2) = -1 - i \cdot \text{Tr}(A) + \det(A) = 0.$$

$$\Rightarrow i^2 - i \cdot \lambda_2 - i \cdot \lambda_1 + \lambda_1 \lambda_2 = -1 - i \cdot \text{Tr}(A) + \underline{\det(A)} = 0.$$

$$\Rightarrow \underline{\det(A)} = 1 + i \cdot \text{Tr}(A)$$

$$\det(A - 2iI_2) = p(2)$$

$$p(2) = (2 - \lambda_1)(2 - \lambda_2) = 4 - 2\lambda_2 - 2\lambda_1 + \lambda_1 \lambda_2$$

$$= 4 - 2(\lambda_1 + \lambda_2) + \lambda_1 \lambda_2$$

$$p(2) = 2^2 - 2 \cdot (a+d) + \underline{ad - bc} = 4 - 2 \underline{\text{Tr}(A)} + \underline{\det A}$$

$$= 4 - 2 \cdot \text{Tr}(A) + 1 + i \cdot \text{Tr}(A)$$

$$= 5 + (i-2) \cdot \text{Tr}(A)$$

$$\det(A) = 1 + i \cdot \text{Tr}(A) \Rightarrow \underline{\text{Tr}(A) = 0}$$

$$\underbrace{\det(A)}_{\in \mathbb{R}} = \underbrace{1}_{\in \mathbb{R}} + i \underbrace{\text{Tr}(A)}_{\in \mathbb{R}} \Rightarrow \det(A - 2iI_2) = 5 + (i-2) \cdot 0 = 5.$$

